

10.4 Solution: (a) By replacing ϵ_r with

$$\epsilon_r + i\frac{\sigma}{\omega\epsilon_0} = \epsilon_r + i\frac{Z_0\sigma}{k},$$

and using Eqs. (10.43) and (10.11), the differential and total scattering cross sections are, respectively,

$$\frac{d\sigma_{sc}}{d\Omega} = \frac{1}{2}k^4a^6(1 + \cos^2\theta) \left| \frac{\epsilon_r - 1}{\epsilon_r + 2} \right|^2 = \frac{1}{2}k^4a^6(1 + \cos^2\theta) \frac{(\epsilon_r - 1)^2 + (Z_0\sigma/k)^2}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2},$$

and

$$\sigma_{sc} = \frac{8\pi}{3}k^4a^6 \frac{(\epsilon_r - 1)^2 + (Z_0\sigma/k)^2}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2}.$$

(b) From Eq. (10.54), the absorbed power is

$$\begin{aligned} P_{\text{abs}} &= -\frac{1}{2\mu_0} \oint_S \text{Re}[\mathbf{E} \times \mathbf{H}^*] \cdot \hat{\mathbf{n}} da = -\frac{1}{2} \text{Re} \int \nabla \cdot (\mathbf{E} \times \mathbf{H}^*) d^3x \\ &= -\frac{1}{2} \text{Re} \int [\mathbf{H}^* \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}^*)] d^3x \\ &= -\frac{1}{2} \text{Re} \int \left[\mathbf{H}^* \cdot \left(-\frac{\partial \mathbf{B}}{\partial t} \right) - \mathbf{E} \cdot \mathbf{J}^* - \mathbf{E} \cdot \frac{\partial \mathbf{D}^*}{\partial t} \right] d^3x \\ &= \frac{1}{2} \text{Re} \int \mathbf{J}^* \cdot \mathbf{E} d^3x + \frac{1}{2} \text{Re} \int \left[\mathbf{H}^* \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \frac{\partial \mathbf{D}^*}{\partial t} \right] d^3x \\ &= \frac{1}{2} \int \sigma |\mathbf{E}|^2 d^3x + \frac{1}{4} \frac{\partial}{\partial t} \int \left(\frac{1}{\mu_0} |\mathbf{B}|^2 + \epsilon_0 |\mathbf{E}|^2 \right) d^3x. \end{aligned}$$

The second term is zero, as the total energy is constant in the sphere. From Eq. (4.53), the electric field inside the sphere is constant and is given by

$$\mathbf{E} = \frac{3}{\epsilon_r + 2} \mathbf{E}_i.$$

Therefore,

$$P_{\text{abs}} = \frac{\sigma}{2} \frac{9}{(\epsilon_r + 2)^2} |\mathbf{E}_i|^2 \cdot \frac{4}{3} \pi R^3 = \frac{6\pi R^3}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2} \sigma |\mathbf{E}_i|^2,$$

and

$$\begin{aligned} \sigma_{\text{abs}} &= \frac{P_{\text{abs}}}{P_i} = \frac{6\sigma R^3}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2} \frac{|\mathbf{E}_i|^2}{|\mathbf{E}_i|^2/(2Z_0)} \\ &= 12\pi R^2 \frac{RZ_0\sigma}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2}. \end{aligned}$$

(c) The scattered electric field is

$$\mathbf{E}_{sc} = \frac{1}{4\pi\epsilon_0} k^2 \frac{e^{ikr}}{r} [(\hat{\mathbf{n}} \times \mathbf{p}) \times \hat{\mathbf{n}}],$$

where

$$\mathbf{p} = 4\pi\epsilon_0 \frac{\epsilon_r - 1}{\epsilon_r + 2} R^3 \mathbf{E}_i.$$

In terms of the incident field,

$$\mathbf{E}_{sc} = \frac{e^{ikr}}{r} E_0 k^2 R^3 \frac{\epsilon_r - 1}{\epsilon_r + 2} \left[(\hat{\mathbf{n}} \times \hat{\mathbf{e}}_0) \times \hat{\mathbf{n}} \right].$$

By definition,

$$\mathbf{f}(\hat{\mathbf{k}}, \hat{\mathbf{k}}_0) = k^2 R^3 \frac{\epsilon_r - 1}{\epsilon_r + 2} \left[(\hat{\mathbf{n}} \times \hat{\mathbf{e}}_0) \times \hat{\mathbf{n}} \right].$$

Using the optical theorem,

$$\begin{aligned} \sigma_t &= \frac{4\pi}{k} \text{Im} \left[\hat{\mathbf{e}}_0^* \cdot \mathbf{f}(\hat{\mathbf{k}} = \hat{\mathbf{k}}_0) \right] \\ &= \frac{4\pi}{k} \text{Im} \left[k^2 R^3 \frac{\epsilon_r - 1}{\epsilon_r + 2} \hat{\mathbf{e}}_0^* \cdot \left((\hat{\mathbf{n}} \times \hat{\mathbf{e}}_0) \times \hat{\mathbf{n}} \right) \right]_{\hat{\mathbf{n}}=\hat{\mathbf{n}}_0} \\ &= 4\pi k R^3 \text{Im} \left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right) \\ &= 4\pi k R^3 \frac{3Z_0\sigma/k}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2} = 12\pi R^2 \frac{RZ_0\sigma}{(\epsilon_r + 2)^2 + (Z_0\sigma/k)^2}. \end{aligned}$$

This only agrees with the absorption cross section, as scattering cross section is of higher order in long-wavelength limit.